

1. Configure VPC peering in cross regions.

Objective

To establish a VPC peering connection between two VPCs located in different AWS regions and configure routing for communication between them.

VPC Details

VPC	AWS Region	CIDR Range
VPC-Mumbai	Mumbai (ap-south-1)	10.0.0.0/16
VPC-Singapore	Singapore (ap-southeast-1)	20.0.0.0/16

Steps Performed

Step 1: Create VPC-Mumbai

1. Logged in to the AWS Management Console.
2. Navigated to VPC → Your VPCs.
3. Clicked Create VPC.
4. Created the VPC with the following configuration:
 - a. Name: VPC-Mumbai
 - b. IPv4 CIDR: 10.0.0.0/16
 - c. Region: Mumbai (ap-south-1)
5. Created the VPC successfully.

Step 2: Create VPC-Singapore

1. Switched the AWS region to Singapore (ap-southeast-1).
2. Navigated to VPC → Your VPCs.
3. Clicked Create VPC.
4. Created the VPC with:
 - a. Name: VPC-Singapore
 - b. IPv4 CIDR: 20.0.0.0/16
5. Created the VPC successfully.

Step 3: Create VPC Peering Connection

1. Navigated to VPC → Peering connections.

2. Clicked Create peering connection.
3. Entered the peering connection name:
 - a. Mumbai-Singapore-Peering
4. Selected VPC-Mumbai as the Requester VPC.
5. Selected Another Region as the acceptor region.
6. Selected Asia Pacific (Singapore) as the acceptor region.
7. Selected VPC-Singapore as the Acceptor VPC.
8. Clicked Create peering connection.

Step 4: Accept the Peering Request

1. Switched to the Singapore (ap-southeast-1) region.
2. Navigated to VPC → Peering connections.
3. Selected Mumbai-Singapore-Peering.
4. Clicked Actions → Accept request.
5. Verified that the peering connection status changed to Active.

Step 5: Configure Mumbai Route Table

The route table associated with VPC-Mumbai was updated to send traffic destined for the Singapore VPC through the peering connection.

Route added:

Destination	Target
20.0.0.0/16	Mumbai-Singapore-Peering (pcx-xxxxxxx)

Step 6: Configure Singapore Route Table

The route table associated with VPC-Singapore was updated to send traffic destined for the Mumbai VPC through the peering connection.

Route added:

Destination	Target
10.0.0.0/16	Mumbai-Singapore-Peering (pcx-xxxxxxx)

Step 7: Verification

Verified the following:

- VPC-Mumbai: 10.0.0.0/16
- VPC-Singapore: 20.0.0.0/16
- Peering connection: Mumbai-Singapore-Peering
- Peering status: Active
- Mumbai route table contains route to 20.0.0.0/16
- Singapore route table contains route to 10.0.0.0/16

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☒ VPC only ☐ VPC and more

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

VPC - Mumbai

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
10.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block ☐ IPAM-allocated IPv6 CIDR block ☐ Amazon-provided IPv6 CIDR block ☐ IPv6 CIDR owned by me

Create VPC [Info](#)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

VPC settings

Resources to create [Info](#)
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Creates a tag with a key of 'Name' and a value that you specify.

VPC - Singapore

IPv4 CIDR block [Info](#)
☒ IPv4 CIDR manual input ☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR
20.0.0.0/16
CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)
☒ No IPv6 CIDR block ☐ IPAM-allocated IPv6 CIDR block ☐ Amazon-provided IPv6 CIDR block ☐ IPv6 CIDR owned by me

aws

Search

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Asia Pacific (Mumbai)

Shaik Asif (331116047691)

Shaik%20Asif

VPC > Peering connections > Create peering connection

Name - optional

Create a tag with a key of 'Name' and a value that you specify.

Mumbai-Singapore-Peering

Select a local VPC to peer with

VPC ID (Requester)

vpc-03f2df163921c75fb (VPC - Mumbai)

VPC CIDRs for vpc-03f2df163921c75fb (VPC - Mumbai)

CIDR	Status	Status reason
10.0.0.0/16	Associated	-

Select another VPC to peer with

Account

☒ My account

☐ Another account

Region

☐ This Region (ap-south-1)

☒ Another Region

Asia Pacific (Singapore) (ap-southeast-1)

VPC ID (Accepter)

vpc-07f6636a1c0b0285c

aws

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Asia Pacific (Singapore)

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VPC > Peering connections

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Peering connections (1/1) info

Find peering connections by attribute or tag

Name	Peering connection ID	Status	Requester VPC	Accepter VPC
-	pcx-02ee5f69327fa1ded	Active	vpc-03f2df163921c75fb	vpc-07f6636a1c0b0285c / VPC ...

pcx-02ee5f69327fa1ded

Details

Requester owner ID

331116047691

Peering connection ID

pcx-02ee5f69327fa1ded

Status

Active

Expiration time

-

Accepter owner ID

331116047691

Requester VPC

vpc-03f2df163921c75fb

Requester CIDRs

10.0.0.0/16

Requester Region

Mumbai (ap-south-1)

VPC Peering connection ARN

arn:aws:ec2:ap-southeast-1:331116047691:vpc-peering-connection/pcx-02ee5f69327fa1ded

Accepter VPC

vpc-07f6636a1c0b0285c / VPC - Singapore

Accepter CIDRs

20.0.0.0/16

Accepter Region

Singapore (ap-southeast-1)

aws

Search

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Asia Pacific (Singapore)

Shaik Asif (331116047691)

Shaik%20Asif

VPC > Route tables > rtb-0e7a8c0b289e57da7

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

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Peering connections

Route servers

Security

Network ACLs

Security groups

rtb-0e7a8c0b289e57da7

Actions

Details

Route table ID

rtb-0e7a8c0b289e57da7

VPC

vpc-07f6636a1c0b0285c | VPC - Singapore

Main

Yes

Owner ID

331116047691

Explicit subnet associations

-

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

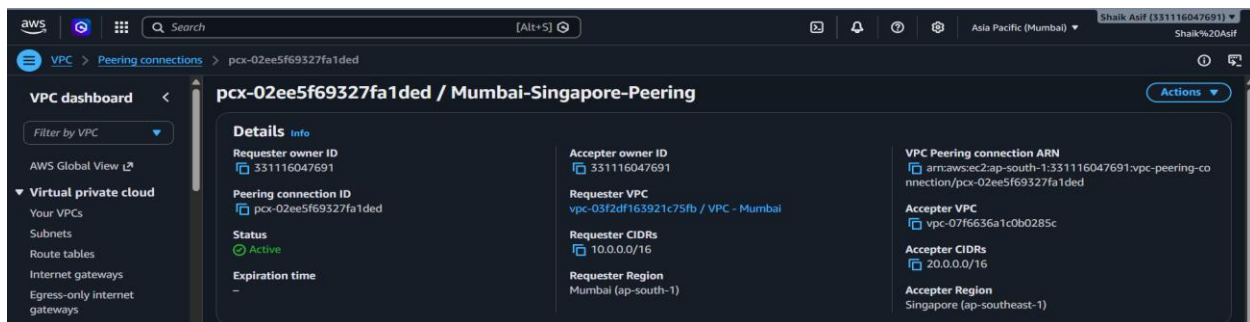
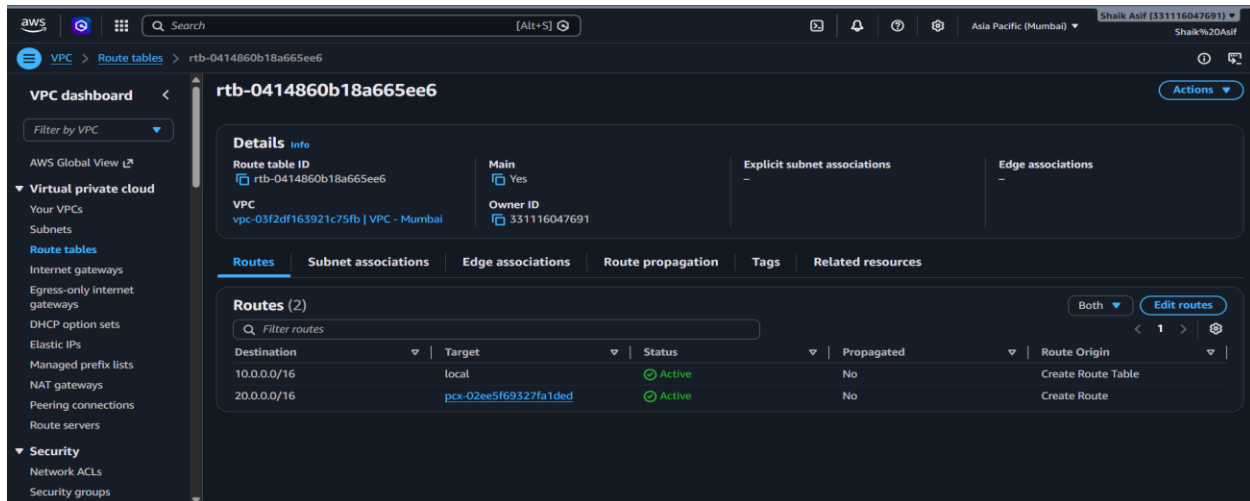
Tags

Related resources

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	pcx-02ee5f69327fa1ded	Active	No	Create Route
20.0.0.0/16	local	Active	No	Create Route Table



Conclusion

Cross-region VPC peering was successfully configured between VPC-Mumbai (10.0.0.0/16) in the Mumbai region and VPC-Singapore (20.0.0.0/16) in the Singapore region. The Mumbai-Singapore-Peering connection was successfully created and accepted, and its status was verified as Active. Routes were configured in both VPC route tables to enable network traffic between the two VPCs through the peering connection.

2. Purchase one domain from GoDaddy.

Objective

To purchase a domain from GoDaddy that can be used for the AWS S3, CloudFront, SSL certificate, and Route 53 configuration.

Domain Purchased

Parameter	Details
Domain Provider	GoDaddy
Domain Name	asifshaik.online
Extension	.online
Expiry Date	07-Sep-2027
Auto-Renew	OFF
Purpose	AWS CloudFront & Route 53 practice

Steps Performed

Step 1: Open GoDaddy

Opened the GoDaddy website and logged in to the GoDaddy account.

Step 2: Search for a Domain

Navigated to the Domains section and searched for a suitable domain name for the AWS practice assignment.

Step 3: Select the Domain

Selected the available domain:

asifshaik.online

Step 4: Add the Domain to Cart

Added asifshaik.online to the shopping cart.

Step 5: Proceed to Checkout

Reviewed the domain registration details and proceeded with the checkout process.

Step 6: Complete Registration

Completed the required GoDaddy registration and payment process.

Step 7: Verify Domain

After successful registration, navigated to:

GoDaddy → Domain Portfolio → asifshaik.online

Step 8: Verify Domain Status

Verified that the domain was successfully registered and is available in the GoDaddy Domain Portfolio.

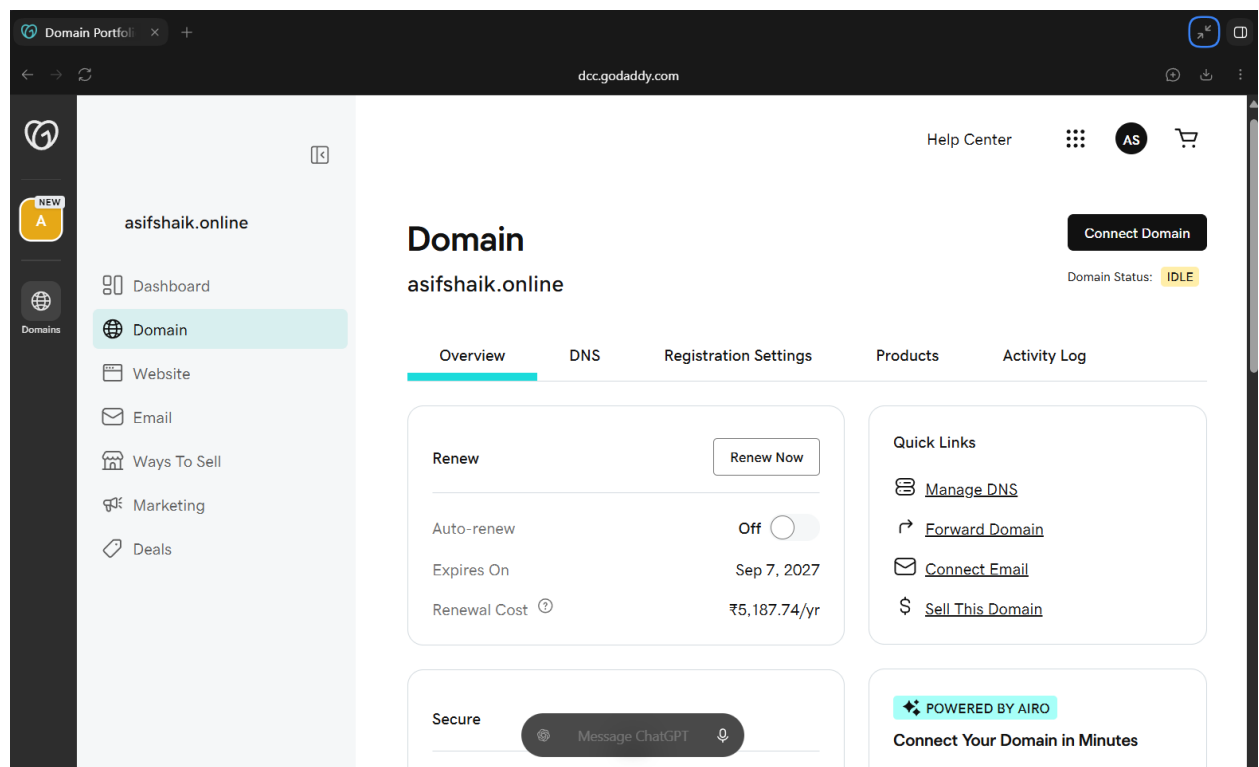
Step 9: Verify Expiration Date

Verified that the domain expiration date is:

07-Sep-2027

Step 10: Disable Auto-Renewal

Verified that Auto-Renew is OFF. This prevents automatic renewal charges after the domain expires.



Conclusion

Successfully purchased and registered the domain asifshaik.online from GoDaddy. The domain is valid until 07-Sep-2027 and is ready to be integrated with AWS Route 53 and CloudFront for the static website deployment. Auto-renewal was kept disabled because the domain is being used for practice purposes.

3. Deploy static website in S3.

Objective

The objective of this task was to deploy a static website using Amazon S3. The website was created using an HTML file, uploaded to an S3 bucket, and configured for static website hosting.

Procedure

Step 1: Create an S3 Bucket

1. Logged in to the AWS Management Console.
2. Opened the Amazon S3 service.
3. Selected Create bucket.
4. Entered a unique name for the S3 bucket.
5. Selected the required AWS Region.
6. Created the bucket.

The S3 bucket acts as the storage location for the static website files.

Step 2: Create the Website

Created an HTML file named index.html.

The webpage was customized for the AWS and DevOps project. It contains information about the project and the AWS services that will be used, including S3, CloudFront, AWS Certificate Manager, and Route 53.

The domain purchased for the project is:

asifshaik.online

Step 3: Upload index.html to the S3 Bucket

1. Opened the newly created S3 bucket.
2. Selected Upload.
3. Added the index.html file.
4. Started the upload.
5. Verified that the file was successfully uploaded to the bucket.

The index.html file serves as the main webpage of the static website.

Step 4: Enable Static Website Hosting

1. Opened the S3 bucket.
2. Navigated to the Properties tab.
3. Located the Static website hosting section.
4. Selected Edit.
5. Enabled static website hosting.
6. Selected Host a static website.
7. Entered index.html as the Index document.
8. Saved the configuration.

After enabling website hosting, Amazon S3 provided a website endpoint that can be used to access the static website.

Step 5: Configure Bucket Access

Configured the S3 bucket permissions required for the website to be accessed through the S3 website endpoint.

A bucket policy was configured to allow the required s3:GetObject permission for the website objects.

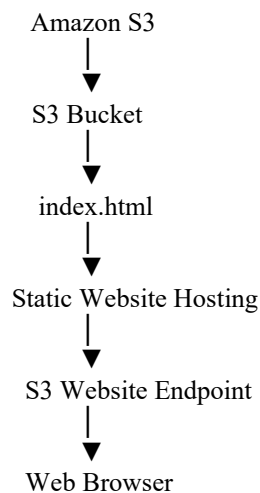
This allows users to retrieve the HTML file from the S3 bucket when accessing the website.

Step 6: Verify the Website

1. Opened the Properties tab of the S3 bucket.
2. Navigated to the Static website hosting section.
3. Copied the S3 website endpoint.
4. Opened the endpoint in a web browser.
5. Verified that the customized website was displayed successfully.

The website was successfully hosted and accessed through Amazon S3.

Architecture



AWS

Search

[Alt+S]

Asia Pacific (Mumbai)

Shaik Asif (331116047691)

Shaik%20Asif

Amazon S3

Buckets

devops-static-website-asif-2026

devops-static-website-asif-2026

Info

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems

Access Points

Objects (0)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name

Type

Last modified

Size

Storage class

AWS

Search

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Asia Pacific (Mumbai)

Shaik Asif (331116047691)

Shaik%20Asif

Amazon S3

Buckets

devops-static-website-asif-2026

index.html

index.html

Info

Properties

Permissions

Versions

Object overview

Owner

3c230a6da1d8b04e69c4d1dc392ca896c9876479133847483429bf37a9182e61

AWS Region

Asia Pacific (Mumbai) ap-south-1

Last modified

September 7, 2026, 17:16:31 (UTC+05:30)

Size

1.0 KB

Type

html

Key

index.html

S3 URI

s3://devops-static-website-asif-2026/index.html

Amazon Resource Name (ARN)

arn:aws:s3:::devops-static-website-asif-2026/index.html

Entity tag (Etag)

d059b8813276e2ec4943e95941de4fad

Object URL

https://devops-static-website-asif-2026.s3.ap-south-1.amazonaws.com/index.html

Static website hosting

Edit

Use this bucket to host a website or redirect requests. [Learn more](#)

We recommend using AWS Amplify Hosting for static website hosting

Deploy a fast, secure, and reliable website quickly with AWS Amplify Hosting. [Learn more about Amplify Hosting](#) or [View your existing Amplify apps](#)

Create Amplify app

S3 static website hosting

Enabled

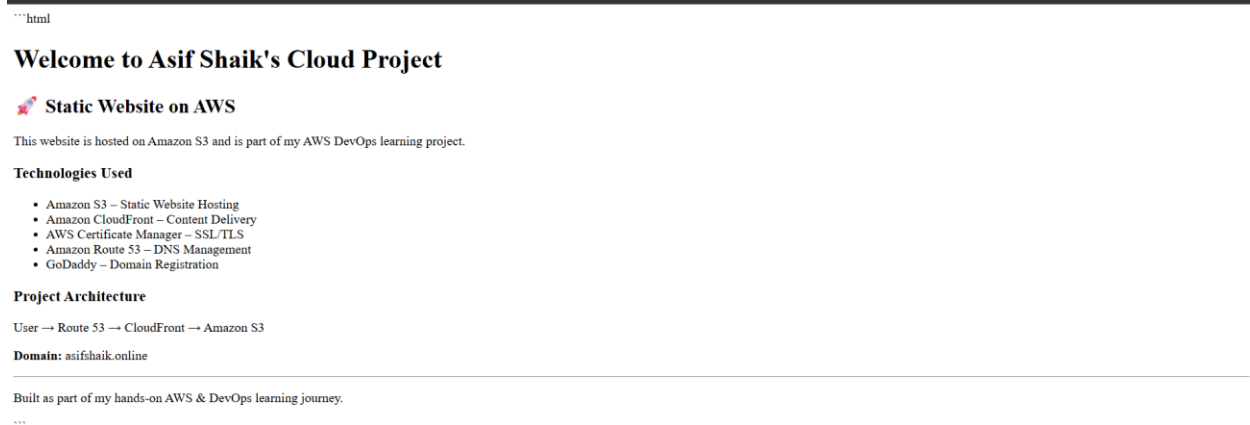
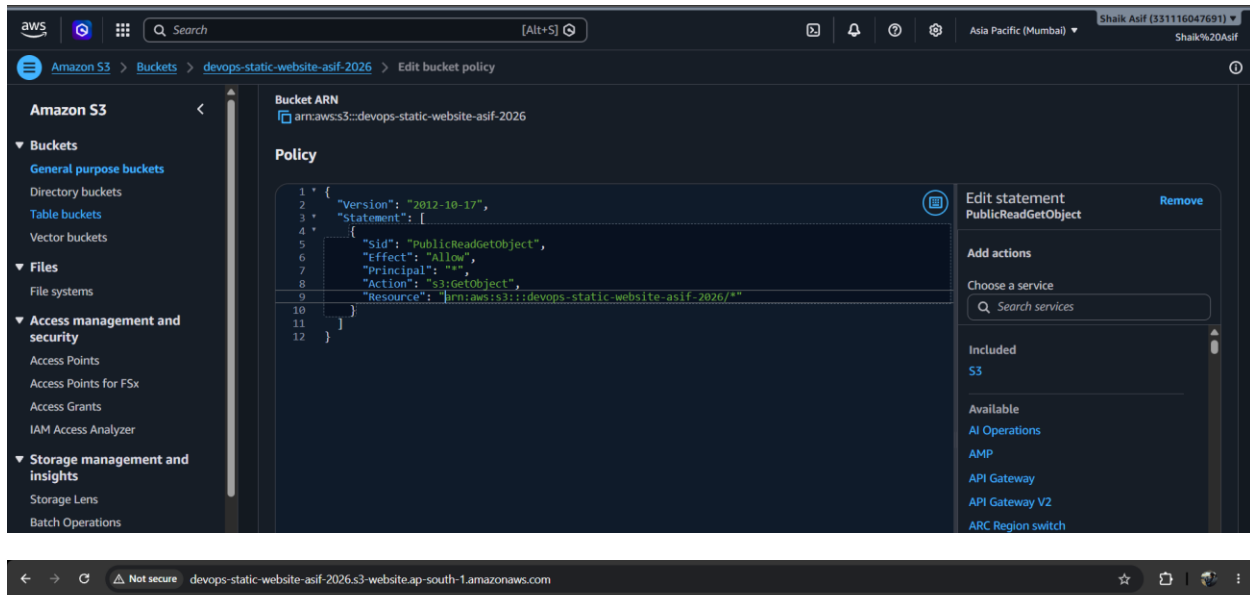
Hosting type

Bucket hosting

Bucket website endpoint

When you configure your bucket as a static website, the website is available at the AWS Region-specific website endpoint of the bucket. [Learn more](#)

<http://devops-static-website-asif-2026.s3-website-ap-south-1.amazonaws.com>



Conclusion

The static website was successfully deployed using Amazon S3. An S3 bucket was created, the customized index.html file was uploaded, static website hosting was enabled, and the required access permissions were configured.

The website was successfully verified through the S3 website endpoint.

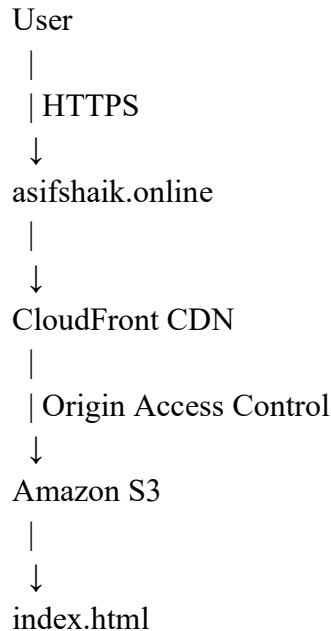
4. Create a CDN and attach one SSL certificate.

1. Objective

The objective of this task was to create an Amazon CloudFront distribution for the static website hosted in an Amazon S3 bucket and secure the website using an AWS Certificate Manager (ACM) SSL/TLS certificate.

CloudFront acts as the CDN layer between the users and the S3 bucket. It improves website delivery by serving content through AWS edge locations and provides HTTPS support using the ACM certificate.

Architecture



2. Prerequisites

Before creating the CloudFront distribution, the following components were already configured:

1. An S3 bucket containing the static website files.
2. The index.html file was uploaded to the S3 bucket.
3. The domain asifshaik.online was purchased through GoDaddy.
4. A Route 53 hosted zone was created for asifshaik.online.
5. GoDaddy nameservers were updated with the Route 53 nameservers.
6. An ACM public SSL certificate was requested for:
 - a. asifshaik.online
 - b. *.asifshaik.online
7. DNS validation for the ACM certificate was completed.
8. The ACM certificate status changed to Issued.
9. The ACM certificate was created in US East (N. Virginia) – us-east-1, which is required when using an ACM certificate with CloudFront.

3. Open Amazon CloudFront

1. Logged in to the AWS Management Console.
2. Searched for CloudFront using the AWS Console search.
3. Opened the Amazon CloudFront service.
4. Navigated to the CloudFront distributions section.
5. Clicked Create distribution.

CloudFront provides the CDN functionality required to distribute the S3 website content to users.

4. Configure the CloudFront Distribution Name

A distribution name was configured to identify the CloudFront distribution.

The name used was:

asif-cloudfront-distribution

This name is used for identifying the distribution within the AWS CloudFront console.

5. Configure the Origin

Origin Domain

The Origin is the location from which CloudFront retrieves the website content.

Since the website was hosted in Amazon S3, the S3 bucket was selected as the CloudFront origin.

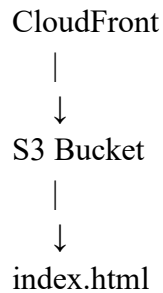
The S3 bucket containing the index.html file was selected under:

Origin domain

The S3 bucket itself was selected rather than the S3 static website endpoint.

Purpose

The configuration establishes the following connection:



CloudFront retrieves the website content from the S3 bucket and delivers it to users.

6. Configure Origin Access Control

Under Origin access, the following option was selected:

Origin access control settings (recommended)

A new Origin Access Control was created with the name:

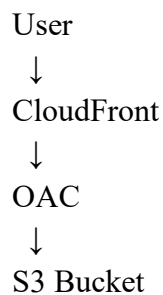
asif-s3-oac

Purpose of Origin Access Control

Origin Access Control (OAC) allows CloudFront to securely access the S3 bucket.

Instead of allowing users to directly access the S3 bucket, the architecture is configured so that users access the website through CloudFront.

The intended flow is:



This provides better control over access to the S3 origin.

7. Configure Default Root Object

The Default root object was configured as:

index.html

Purpose

The default root object tells CloudFront which file to return when a user accesses the root of the website.

For example, when a user visits:

<https://asifshaik.online/>

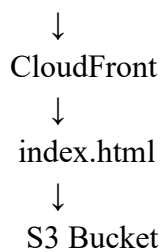
CloudFront uses:

index.html

as the default page.

Therefore, the request flow becomes:

<https://asifshaik.online/>



8. Configure Alternate Domain Name

Under Alternate domain name (CNAME), the custom domain was added:

asifshaik.online

This allows CloudFront to respond to requests made using the custom domain instead of only using the CloudFront-generated domain.

The custom domain is:

asifshaik.online

The CloudFront distribution domain generated by AWS is:

d2d8yyjjy5cuvp.cloudfront.net

The custom domain will later be mapped to this CloudFront distribution using Route 53.

9. Attach the ACM SSL Certificate

Under Custom SSL certificate, the previously created ACM certificate was selected.

The certificate was issued for:

asifshaik.online
*.asifshaik.online

The certificate status was:

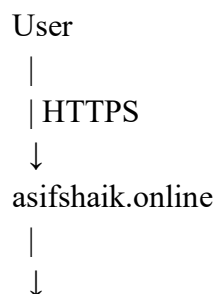
Issued

Purpose of the SSL certificate

The ACM certificate provides HTTPS encryption for the website.

Without the certificate, the website would not be able to use the custom domain with HTTPS through CloudFront.

The intended secure connection is:



CloudFront



S3

The ACM certificate encrypts the connection between the user and CloudFront.

10. Configure Viewer Protocol Policy

The Viewer protocol policy was configured as:

Redirect HTTP to HTTPS

Purpose

This configuration automatically redirects users from HTTP to HTTPS.

For example:

<http://asifshaik.online>

will be redirected to:

<https://asifshaik.online>

This ensures that users access the website using a secure HTTPS connection.

11. Configure Web Application Firewall

Under the CloudFront security settings, the following option was selected:

Do not enable security protections

AWS WAF was not enabled because it was not required for the assignment.

The assignment required:

- CloudFront CDN
- SSL certificate

- S3 website delivery
- Route 53 domain mapping

WAF was therefore left disabled to avoid adding an unnecessary service and additional cost.

12. Review the Configuration

Before creating the distribution, the configuration was reviewed.

The important configuration values were:

Configuration	Value
Distribution Name	asif-cloudfront-distribution
Origin	S3 bucket
Origin Access	Origin Access Control
OAC Name	asif-s3-oac
Default Root Object	index.html
Alternate Domain	asifshaik.online
SSL Certificate	ACM Issued Certificate
SSL Domains	asifshaik.online, *.asifshaik.online
Viewer Protocol Policy	Redirect HTTP to HTTPS
WAF	Not enabled

13. Create the CloudFront Distribution

After reviewing the configuration:

1. Clicked Create distribution.
2. AWS started creating the CloudFront distribution.
3. The distribution initially showed a deployment/creation status.
4. CloudFront generated a distribution domain name.

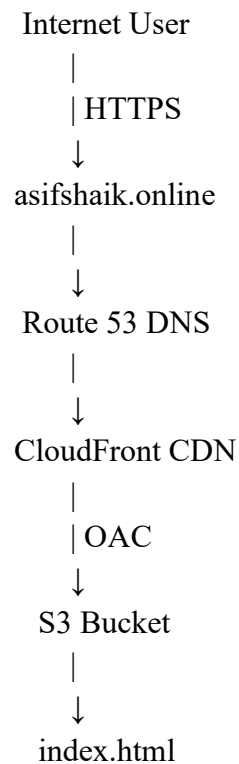
The CloudFront distribution domain name generated was:

d2d8yyjjy5cuvp.cloudfront.net

This domain can be used to access the CloudFront distribution directly.

14. CloudFront Distribution Flow

After successful creation, the website architecture is:



CloudFront receives the user's request, retrieves the required website content from S3, and delivers the content to the user.

15. CloudFront Distribution Details

The CloudFront distribution was successfully created.

CloudFront Distribution Domain

d2d8yyjjy5cuvp.cloudfront.net

Custom Domain

asifshaik.online

Origin

Amazon S3 Bucket

Default Page

index.html

SSL/TLS

AWS Certificate Manager (ACM)

HTTPS Configuration

Redirect HTTP to HTTPS

16. Task Completion Status

The CloudFront CDN was successfully created and configured with the ACM SSL certificate.

The following components have been completed:

- S3 website hosting — Completed
- ACM SSL certificate — Issued
- CloudFront distribution — Created
- S3 configured as CloudFront origin — Completed
- Origin Access Control — Configured
- Custom domain — Added to CloudFront
- SSL certificate — Attached
- HTTP to HTTPS redirection — Configured
- WAF — Not enabled

Route 53

Hosted zones

asifshaik.online

Public

asifshaik.online

Info

Delete zone

Test record

Configure query logging

Hosted zone details

Edit hosted zone

Records (3)

Accelerated recovery

DNSSEC signing

Hosted zone tags (0)

Records (3)

Info

Delete record

Import zone file

Create record

Automatic mode is the current search behavior optimized for best filter results. To change modes go to settings.

Filter records by property or value

Type

Routing p...

Alias

< 1 >

Filter icon

Record ...

Type

Routing...

Differ...

Alias

Value/Route traffic to

asifshaik...

NS

Simple

-

No

ns-864.awsdns-44.net.
ns-1693.awsdns-19.co.uk
ns-73.awsdns-09.com.
ns-1311.awsdns-35.org.

asifshaik...

SOA

Simple

-

No

ns-864.awsdns-44.net. as

_7228911...

CNAME

Simple

-

No

_d847d7fb56b8581e80e

AWS Certificate Manager

Certificates

e0b25058-9c59-480c-9591-1d87df7b5f78

Delete

Certificate status

Identifier

e0b25058-9c59-480c-9591-1d87df7b5f78

Status

Issued

ARN

arn:aws:acm:us-east-1:331116047691:certificate/e0b25058-9c59-480c-9591-1d87df7b5f78

Key source

AWS managed

Type

Amazon Issued

Domains (2)

Create records in Route 53

Export to CSV

Domain

Status

Renewal status

Type

CNAME name

asifshaik.online

Success

-

CNAME

_7228911c2fe570ae8abe9c0b91a38536.asifshai

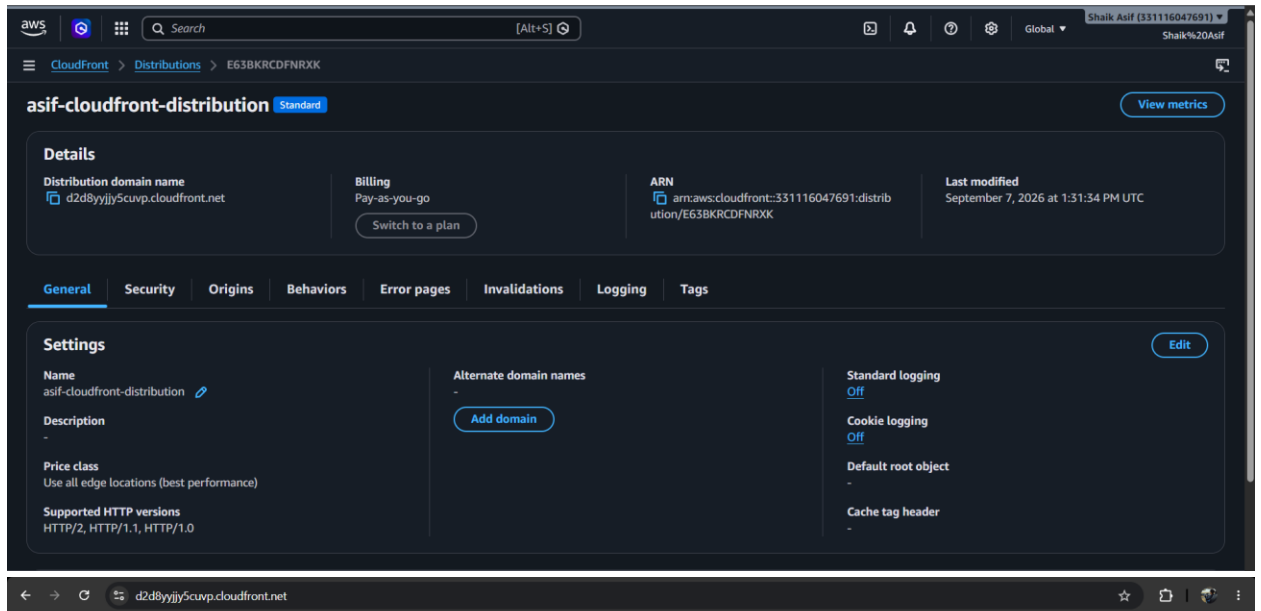
*.asifshaik.online

Success

-

CNAME

_7228911c2fe570ae8abe9c0b91a38536.asifshai



***html

Welcome to Asif Shaik's Cloud Project

Static Website on AWS

This website is hosted on Amazon S3 and is part of my AWS DevOps learning project.

Technologies Used

- Amazon S3 – Static Website Hosting
- Amazon CloudFront – Content Delivery
- AWS Certificate Manager – SSL/TLS
- Amazon Route 53 – DNS Management
- GoDaddy – Domain Registration

Project Architecture

User → Route 53 → CloudFront → Amazon S3

Domain: asifshaik.online

Built as part of my hands-on AWS & DevOps learning journey.

...

Conclusion

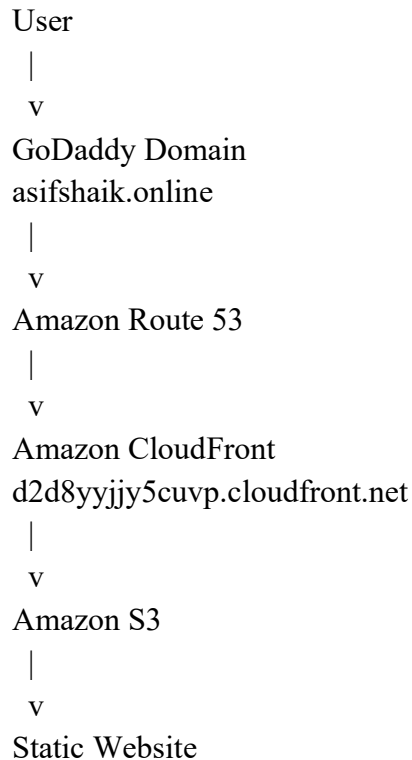
The GoDaddy domain asifshaik.online was successfully delegated to Amazon Route 53 by replacing the GoDaddy nameservers with the Route 53 nameservers. Route 53 was then configured to route the domain to the CloudFront distribution, allowing the S3-hosted static website to be accessed through the custom domain with HTTPS.

5. Create a Route 53 hosted zone and map the domain with the CDN.

Objective

To deploy a static website using Amazon S3, distribute it through Amazon CloudFront, secure it using an AWS Certificate Manager (ACM) SSL/TLS certificate, and map a GoDaddy domain using Amazon Route 53.

Architecture



1. Purchase Domain from GoDaddy

1. Logged in to the GoDaddy account.
2. Searched for the required domain.
3. Purchased the domain:

asifshaik.online

4. The domain was used as the custom domain for the AWS-hosted website.

Evidence: Screenshot of the GoDaddy domain showing asifshaik.online.

2. Create an S3 Bucket

1. Opened the AWS Management Console.

2. Navigated to:

AWS Console → S3 → Create bucket

3. Created an S3 bucket for hosting the website.

4. Uploaded the static website files, including:
index.html

5. Configured the bucket for static website hosting as required.

The website content included information about the AWS project and its architecture.

Evidence: Screenshot showing the S3 bucket and uploaded website files.

3. Deploy the Static Website

1. Uploaded the website's index.html file to the S3 bucket.

2. Configured the website's index document as:
index.html

3. Verified that the website content was available from the S3 origin.

The website displayed content such as:

Welcome to Asif Shaik's Cloud Project

Static Website on AWS

Technologies Used:

Amazon S3

CloudFront

ACM

Route 53

GoDaddy

4. Request SSL/TLS Certificate using ACM

1. Opened:

AWS Console → Certificate Manager (ACM)

2. Requested a public certificate.
3. Added the following domain names:

asifshaik.online

*.asifshaik.online

The wildcard certificate allows HTTPS for subdomains such as:

www.asifshaik.online

4. Selected DNS validation.
5. Added/completed the required DNS validation records.
6. Waited until the certificate status became:

Success / Issued

Certificate details

Domain 1: asifshaik.online

Domain 2: *.asifshaik.online

Key type: RSA 2048

Validation: DNS

Status: Issued

Important: The ACM certificate used with CloudFront was created in the US East (N. Virginia) – us-east-1 region.

Evidence: Screenshot showing the issued ACM certificate.

5. Create CloudFront Distribution

1. Opened:

AWS Console → CloudFront

2. Selected Create distribution.
3. Configured the S3 bucket as the origin.
4. Configured the default behavior to serve the website through CloudFront.
5. Configured the default/root object as:

index.html

6. Enabled HTTPS support.
7. Selected the ACM certificate created earlier.
8. Added the custom domain under Alternate domain names (CNAMEs):
asifshaik.online
9. Completed the CloudFront distribution creation.

After deployment, CloudFront generated the distribution domain:

d2d8yyjjy5cuvp.cloudfront.net

6. Verify CloudFront

After the CloudFront distribution was deployed, accessed:

<https://d2d8yyjjy5cuvp.cloudfront.net>

The website loaded successfully.

This confirmed that:

- S3 origin was working.
- CloudFront distribution was working.
- Website content was being delivered through CloudFront.

Evidence: Screenshot of the website opened using the CloudFront URL.

7. Create Route 53 Hosted Zone

1. Opened:
AWS Console → Route 53
2. Created a Hosted Zone for:
asifshaik.online

3. Route 53 automatically created the required NS and SOA records.

Evidence: Screenshot of the Route 53 hosted zone.

8. Configure Domain DNS in Route 53

Created an A record for the root domain.

Record configuration

Name: asifshaik.online

Type: A

Routing Policy: Simple

Alias: Yes

Route traffic to: CloudFront distribution

Target: d2d8yyjjy5cuvp.cloudfront.net

The Alias record routes requests for the custom domain to the CloudFront distribution.

Evidence: Screenshot showing the Route 53 A Alias record pointing to CloudFront.

9. Verify Custom Domain

After the CloudFront configuration and DNS changes were fully deployed, opened:

<https://asifshaik.online>

The website loaded successfully.

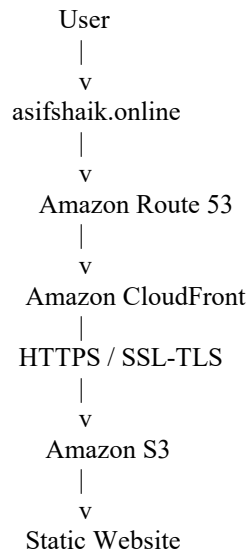
Also verified the CloudFront URL:

<https://d2d8yyjjy5cuvp.cloudfront.net>

Both URLs successfully displayed the static website.

10. Final Architecture Verification

The complete request flow is:



Technologies Used

Technology	Purpose
GoDaddy	Domain registration
Amazon Route 53	DNS management
Amazon S3	Static website hosting/storage
Amazon CloudFront	CDN and content delivery
AWS Certificate Manager (ACM)	SSL/TLS certificate
HTTPS	Secure communication

aws

Search

[Alt+S]

Shaik Asif (331116047691)

Shaik%20Asif

Route 53

Hosted zones

asifshaik.online

Create record

Create record

Info

Quick create record

Switch to wizard

Record 1

Delete

Record name

Info

subdomain

asifshaik.online

Record type

Info

A – Routes traffic to an IPv4 address and some AWS resources

Alias

Route traffic to

Info

Alias to CloudFront distribution

US East (N. Virginia)

An alias to a CloudFront distribution and another record in the same hosted zone are global and available only in US East (N. Virginia).

d2d8yjjy5cuvp.cloudfront.net

Routing policy

Info

Simple routing

Evaluate target health

No

Add another record

Asif Shaik

Dashboard

Domain

Website

Email

Store

Appointments

Marketing

Conversations

Customers

Users

Deals

Domain

asifshaik.online

Connect Domain

Overview

DNS

Registration Settings

Products

Activity Log

DNS Records

Forwarding

Nameservers

Hostnames

DNSSEC

Nameservers

determine where your DNS is hosted and where you add, edit or delete your DNS records.

Using custom nameservers

Change Nameservers

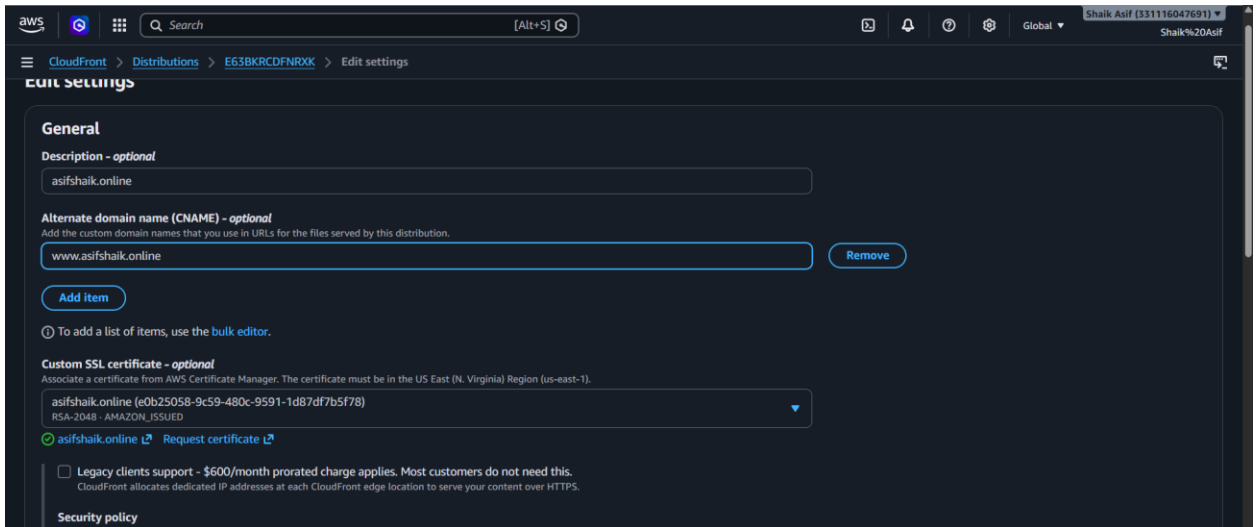
Nameservers

ns-864.awsdns-44.net

ns-1693.awsdns-19.co.uk

ns-73.awsdns-09.com

ns-1311.awsdns-35.org



```html

## Welcome to Asif Shaik's Cloud Project

### Static Website on AWS

This website is hosted on Amazon S3 and is part of my AWS DevOps learning project.

#### Technologies Used

- Amazon S3 – Static Website Hosting
- Amazon CloudFront – Content Delivery
- AWS Certificate Manager – SSL/TLS
- Amazon Route 53 – DNS Management
- GoDaddy – Domain Registration

#### Project Architecture

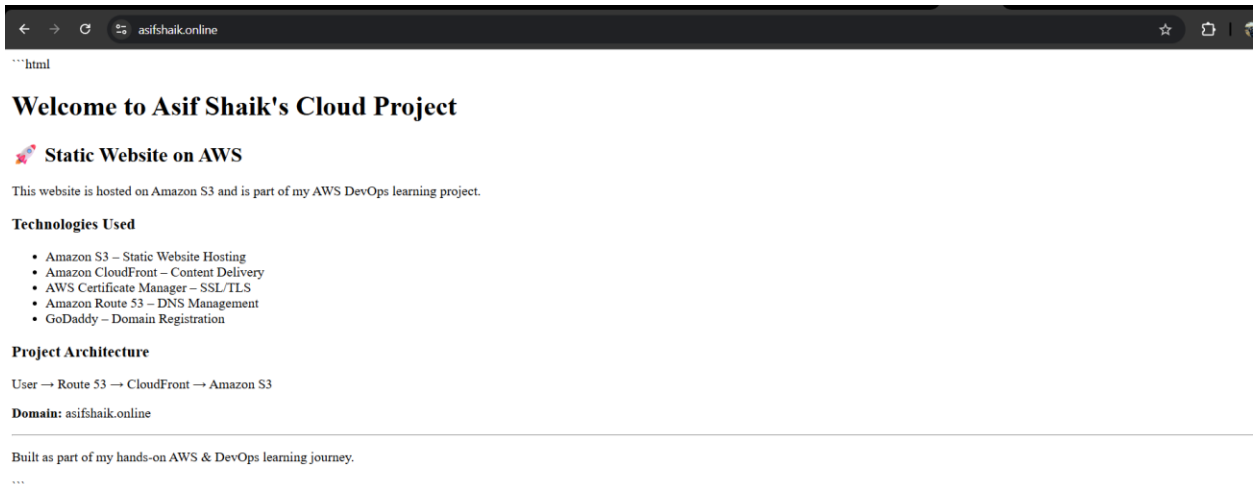
User → Route 53 → CloudFront → Amazon S3

**Domain:** asifshaik.online

---

Built as part of my hands-on AWS & DevOps learning journey.

```



Conclusion

The static website was successfully deployed using Amazon S3 and distributed globally using Amazon CloudFront. An SSL/TLS certificate was successfully created and validated through AWS Certificate Manager (ACM) and associated with the CloudFront distribution.

The domain asifshaik.online, purchased through GoDaddy, was integrated with Amazon Route 53. An A Alias record was configured in Route 53 to route traffic from the custom domain to the CloudFront distribution.

The deployment was successfully verified using both:

<https://asifshaik.online>

and

<https://d2d8yyjy5cuvp.cloudfront.net>

Both URLs successfully displayed the static website over HTTPS.

Final Result

S3 → CloudFront → ACM/HTTPS → Route 53 → GoDaddy Domain configuration was successfully completed and verified.

6. Update the index.html in the S3 bucket and ensure the updated file is accessible using the domain name.

Objective

To update the existing static website hosted in Amazon S3 and ensure that the updated website content is accessible through the custom domain asifshaik.online using CloudFront.

Step 1: Update the index.html File

1. Opened the existing index.html file.
2. Replaced the previous AWS project content with a realistic business website for:

Shukur & Sons Vulcanizing Works

3. Updated the website to include:
 - a. Business name
 - b. Owner name: S. A. Shukur
 - c. About the business
 - d. Vulcanizing services
 - e. Tyre puncture repair
 - f. Tyre services
 - g. Wheel services
 - h. Contact section
4. Saved the updated file as:
index.html

Step 2: Open the S3 Bucket

1. Logged in to the AWS Management Console.
2. Navigated to:
AWS Console → S3
3. Opened the S3 bucket used for the static website.
4. Located the existing:
index.html

Step 3: Upload the Updated index.html

1. Selected the existing website files.
2. Uploaded the newly updated index.html.
3. The new file replaced the previous version.
4. Verified that the updated index.html was present in the S3 bucket.
5. Checked the Last modified time to confirm that the latest file was uploaded.

Evidence: Screenshot showing the updated index.html in the S3 bucket.

Step 4: Create CloudFront Cache Invalidation

Because CloudFront can cache the previous version of the website, the cache was invalidated after updating the S3 file.

1. Navigated to:
AWS Console → CloudFront
2. Selected the existing CloudFront distribution.
3. Opened:
Invalidations
4. Selected Create invalidation.
5. Entered:

/*

6. Created the invalidation.
7. Waited until the invalidation status became:
Completed

This ensured that CloudFront fetched the latest index.html from the S3 origin.

Evidence: Screenshot showing the CloudFront invalidation with status Completed.

Step 5: Verify the CloudFront URL

Opened the CloudFront distribution URL:

<https://d2d8yyjy5cuvp.cloudfront.net>

Verified that the updated Shukur & Sons Vulcanizing Works website was displayed.

Step 6: Verify the Custom Domain

Opened:

<https://asifshaik.online>

The updated website was successfully displayed through the custom domain.

The website showed:

Shukur & Sons
Vulcanizing Works

Owner: S. A. Shukur

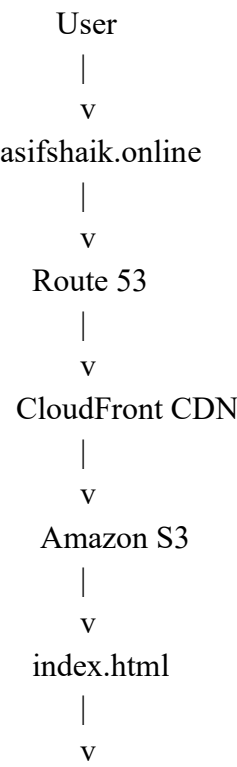
along with the business information and services.

Evidence: Screenshot showing the updated website through <https://asifshaik.online>.

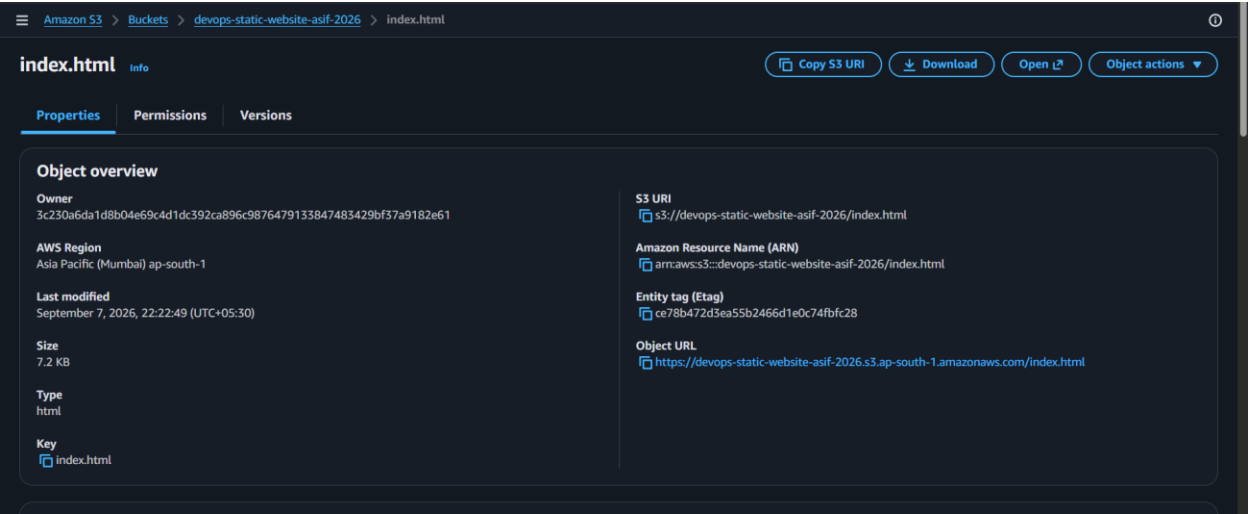
Verification

Component	Result
Updated index.html	✓ Completed
S3 upload	✓ Successful
CloudFront cache invalidation	✓ Completed
CloudFront URL	✓ Updated website accessible
Custom domain	✓ Updated website accessible
HTTPS	✓ Working

Final Architecture



Shukur & Sons Vulcanizing Works
Owner: S. A. Shukur



CloudFront

Distributions

E63BKRCDFNRXX

Create invalidation

CloudFront

Distributions

Policies

Functions

Static IPs

VPC origins

SaaS

Multi-tenant distributions

Distribution tenants

Telemetry

Monitoring

Alarms

Logs

Reports & analytics

Cache statistics

Popular objects

Top referers

Usage

Viewers

Create invalidation

Choose which cached resources will be invalidated so that endpoints will load them again.

Object paths

Selection method

By paths

Specify the paths of the objects to invalidate. You may use wildcards.

By cache tags

Specify objects to invalidate based on the cache tags set in an HTTP header.

Object paths to invalidate

Info

Put each path to invalidate on a separate line. Wildcards (*) are only valid at the end of the path.

/

Cancel

Create invalidation

CloudFront

Distributions

E63BKRCDFNRXX

IVKILKZDASZ2YX4XZL24I7EY6

CloudFront

Distributions

Policies

Functions

Static IPs

VPC origins

SaaS

Multi-tenant distributions

Distribution tenants

Telemetry

Monitoring

Alarms

Logs

Reports & analytics

Cache statistics

Popular objects

Top referers

Usage

Viewers

Invalidation details

Date created

September 7, 2026 at 4:57:04 PM UTC

Status

Completed

Object paths

/

Copy to new

← → ↻ asifshaik.online

Shukur & Sons Vulcanizing Works

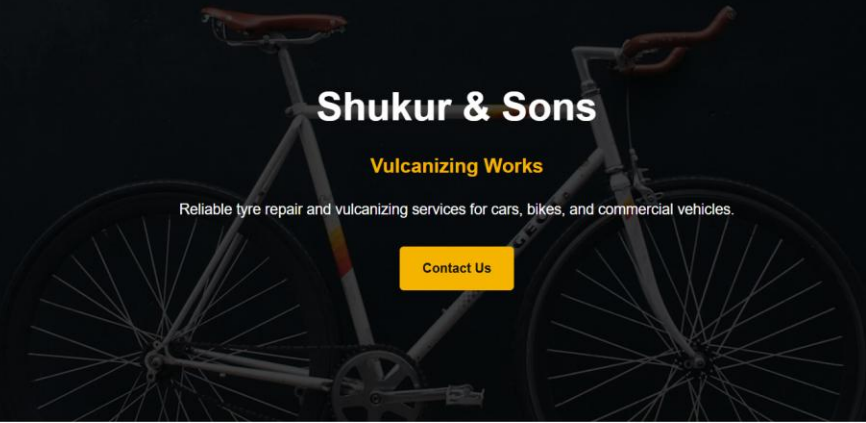
Home About Services Contact

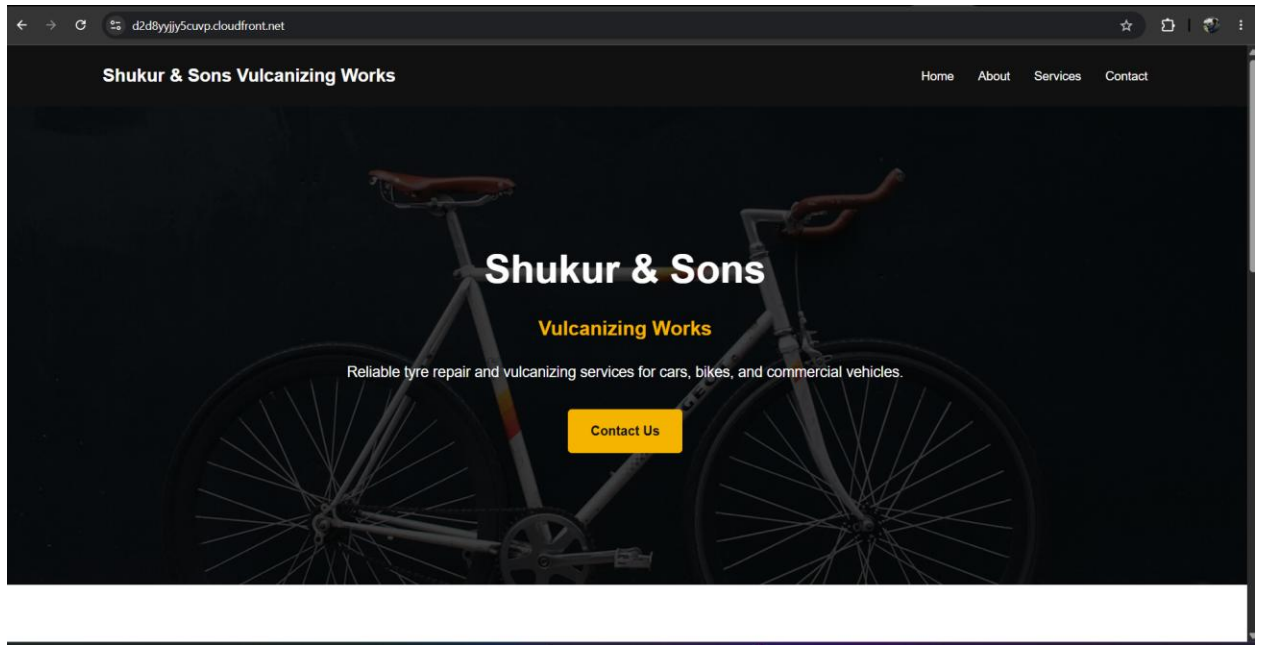
Shukur & Sons

Vulcanizing Works

Reliable tyre repair and vulcanizing services for cars, bikes, and commercial vehicles.

Contact Us





Conclusion

The existing index.html file was successfully updated with the new Shukur & Sons Vulcanizing Works business website owned by S. A. Shukur. The updated file was uploaded to the Amazon S3 bucket hosting the static website.

A CloudFront cache invalidation (/*) was performed to remove the previously cached version and ensure that CloudFront retrieved the latest content from S3.

The updated website was successfully verified through both the CloudFront distribution URL and the custom domain:

<https://d2d8yyjy5cuvp.cloudfront.net><https://asifshaik.online>

Therefore, the S3, CloudFront, Route 53, ACM, and custom-domain configuration successfully delivered the updated website over HTTPS.

Share the domain name in Slack to test the connectivity.

Domain for connectivity testing:

<https://asifshaik.online>